Ultra-high dimensional variable selection for causal mediation analysis

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March 2024

1 Introduction

4.1.3 Variable selection for high-dimensional causal mediation analysis [MSc 3–Malek Achour and BSc 3]: Over the past three decades, variable selection for treatment effect estimation has received considerable attention. However, variable selection in causal mediation analysis is still an emerging topic [2,5,9]. Recently, Ye et al. [9] used the marginal structural models (MSM) to propose a variable selection for causal mediation analysis. However, the adaptive weights used in the application of OAL [7] are the same for both PS models (treatment and mediator), and are specified using an outcome model that conditions on the covariates (X), treatment (A) and mediator (M). As discussed in [2] (page 91) and [5], this strategy can lead to an exclusion of treatment-mediator confounders (C_{AM}) which is undesirable since confounders (C_{AM} are required for unbiased effect estimation. Their approach is intriguing but, some questions remain regarding its functioning and empirical performance. While Ye et al. [9]'s adaptive weights building strategy appears reasonable at first sight, it is hypothesized it could cause OAL to exclude covariates affecting the outcome indirectly through the mediator (such as C_{AM} confounders). Second, although these authors did investigate which sets of covariates should be included in the PS models for calculating the weights, the insight is only partial as the simulation setting only considered one confounding variable for the treatment, mediator and outcome altogether. This is particularly a concern given above remark on the possible exclusion of C_{AM} confounders. In the **short-term** goal, we will propose a general variable selection for high-dimensional causal mediation analysis using MSM that can take into account all four types of confounders, C_{AY}, C_{AM}, C_{MY} and C_{AMY}. **We now provide some**

details on how the method will be developed. First, we will use GOAL [4] for both PS models (treatment and mediator) rather than OAL to deal with high-dimensional data. Second, for the treatment PS model, we will construct the adaptive weights in the application of GOAL [4] using the outcome model that conditions on X and A. Third, for the mediator PS model, we will build the adaptive weights in the application of GOAL [4] using the outcome model that conditions on X, A and M. This new strategy will allow the selection of all four types of confounders (C_{AY}, C_{AM}, C_{MY} and C_{AMY}) which are required for unbiased effect estimation. We will compare our proposed method with the alternative approaches proposed in [5, 9]. **HPQ involvement**: **MSc 3–Malek Achour (MSc at UMoncton under my supervision-Winter 2026)** will work on this project, while **BSc 3** will help implementing the methods in an `R` package.

3. Methods

Algorithm

Given original data $(\mathbf{X}, A, M, Y)$;

Compute $(\hat{\beta}_A^{ols}, \hat{\beta}^{ols}) = \arg\min_{(\beta_A,\beta)} \|Y - \beta_A A - \mathbf{X}\beta\|_2^2$;

Compute $(\tilde{\beta}_A^{ols}, \tilde{\beta}_M^{ols}, \tilde{\beta}^{ols}) = \arg\min_{(\beta_A,\beta_M,\beta)} \|Y - \beta_A A - \beta_M M - \mathbf{X}\beta\|_2^2$;

Set $\hat{w}_j = \left|\hat{\beta}_j^{ols}\right|^{-\gamma}$ with $\gamma > 1,\ j = 1, 2, \ldots, p$;

Set $\tilde{w}_j = \left|\tilde{\beta}_j^{ols}\right|^{-\gamma}$ with $\gamma > 1,\ j = 1, 2, \ldots, p$;

Set $\mathbf{X}^* = \begin{pmatrix} \mathbf{X} \\ \sqrt{\lambda_2}\mathbf{I}_p \end{pmatrix}$ and $A^* = \begin{pmatrix} A \\ 0_p \end{pmatrix}$;

Treatment PS model

a: Call **OAL**

$$\hat{\alpha} = \arg\min_{\alpha} \left[\sum_{i=1}^{n} \left\{ -a_i(x_i^T\alpha) + \log\left(1 + e^{x_i^T\alpha}\right) \right\} + \lambda \sum_{j=1}^{p} \hat{w}_j |\alpha_j| \right];$$

b: Call **GOAL**

$$\hat{\alpha} = \arg\min_{\alpha} \left[\sum_{i=1}^{n+p} \left\{ -a_i^*(x_i^{*T}\alpha) + \log\left(1 + e^{x_i^{*T}\alpha}\right) \right\} + \lambda_1 \sum_{j=1}^{p} \hat{w}_j |\alpha_j| \right];$$

Mediator PS model

a: Call **OAL**

$$(\tilde{\alpha}_A, \tilde{\alpha}) = \arg\min_{\alpha_A,\alpha} \left[\sum_{i=1}^{n} \left\{ -m_i(a_i\alpha_A + x_i^T\alpha) + \log\left(1 + e^{a_i\alpha_A + x_i^T\alpha}\right) \right\} + \lambda_M(\alpha_A + \sum_{j=1}^{p} \tilde{w}_j |\alpha_j|) \right];$$

b: Call **GOAL**

$$(\tilde{\alpha}_A, \tilde{\alpha}) = \arg\min_{\alpha_A,\alpha} \left[\sum_{i=1}^{n} \left\{ -m_i(a_i^*\alpha_A + x_i^{*T}\alpha) + \log\left(1 + e^{a_i^*\alpha_A + x_i^{*T}\alpha}\right) \right\} + \lambda_M(\alpha_A + \sum_{j=1}^{p} \tilde{w}_j |\alpha_j|) \right];$$

Follow Lange et al. (2012) procedure to estimate NDE and NIE (Coming soon)

4. Simulation study

4. 1. Case 1: Ye et al. 2021

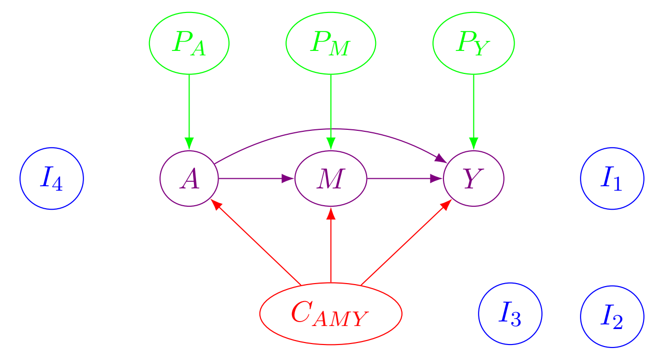

Figure 1: Hypothesized causal mediation graph.

The variables in Figure 1 are the covariates $X = (C_{AMY}, P_Y, P_A, P_M, I_S)$, exposure (A), mediator (M) and outcome (Y). The baseline covariates (X) include exposure-outcome-mediator confounders (C_{AYM}), pure predictors of outcome (P_Y), pure predictors of exposure (P_A), pure predictors of mediator (P_M) and spurious covariates $(I_S = (I_1, I_2, I_3, I_4))$.

4. 2. Case 2: Common causal mediation structure

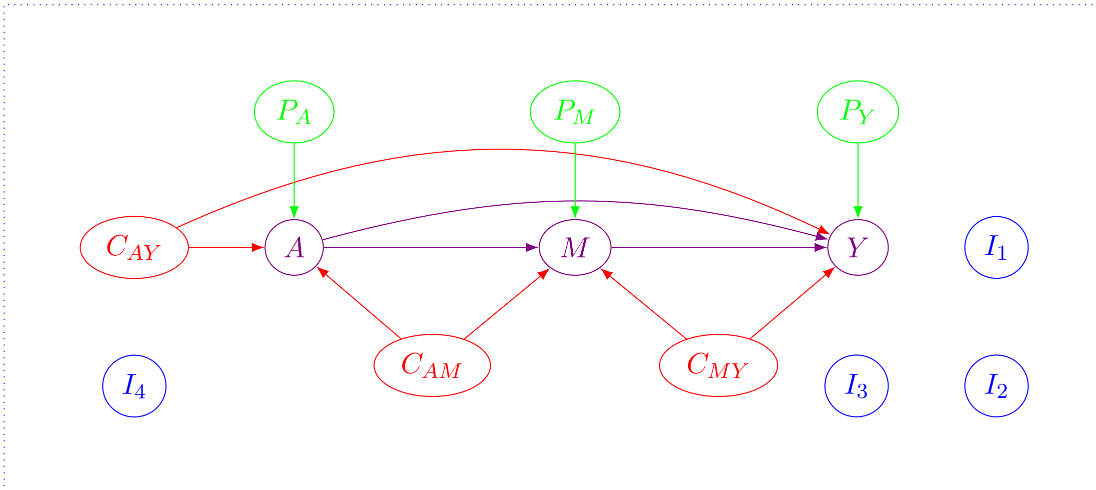

Figure 2: Hypothesized causal mediation graph.

The variables in Figure 1 are the covariates $X = (C_{AY}, C_{MY}, C_{AM}, P_Y, P_A, P_M, I_S)$, exposure ($A$), mediator ($M$) and outcome ($Y$). The baseline covariates (X) include exposure-outcome confounders (C_{AY}), mediator-outcome confounders (C_{MY}), exposure-mediator confounders (C_{AM}), pure predictors of outcome (P_Y), pure predictors of exposure (P_A), pure predictors of mediator (P_M) and spurious covariates ($I_S = (I_1, I_2, I_3, I_4)$).

5. Real data application: Radiomics data

6. Discussion

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